Assignment Problem – Python Programs

This document contains 10 Assignment Problems solved using Python. Each problem includes Aim, Procedure, Python Code, Output, and a detailed explanation of each line of code.

Problem 1

AIM:

To solve Assignment Problem 1 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```

Problem 2

AIM:

To solve Assignment Problem 2 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```

Problem 3

AIM:

To solve Assignment Problem 3 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```

Problem 4

AIM:

To solve Assignment Problem 4 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3)
Minimum Assignment Cost: 13
```

Problem 5

AIM:

To solve Assignment Problem 5 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```

Problem 6

AIM:

To solve Assignment Problem 6 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```

Problem 7

AIM:

To solve Assignment Problem 7 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```

Problem 8

AIM:

To solve Assignment Problem 8 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```

Problem 9

AIM:

To solve Assignment Problem 9 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```

Problem 10

AIM:

To solve Assignment Problem 10 using Python by minimizing the total cost using the Hungarian Method (Linear Assignment).

PYTHON CODE:

```
import numpy as np
from scipy.optimize import linear_sum_assignment

# Step 1: Define the cost matrix
cost_matrix = np.array([[9, 2, 7, 8],
                        [6, 4, 3, 7],
                        [5, 8, 1, 8],
                        [7, 6, 9, 4]])

# Step 2: Apply Hungarian Algorithm
row_ind, col_ind = linear_sum_assignment(cost_matrix)

# Step 3: Calculate minimum cost
min_cost = cost_matrix[row_ind, col_ind].sum()

# Step 4: Display results
print("Optimal Assignments:", list(zip(row_ind, col_ind)))
print("Minimum Assignment Cost:", min_cost)
```

OUTPUT:

```
Optimal Assignments: [(np.int64(0), np.int64(1)), (np.int64(1), np.int64(0)), (np.int64(2), np.int64(2)), (np.int64(3), np.int64(3))]
Minimum Assignment Cost: 13
```